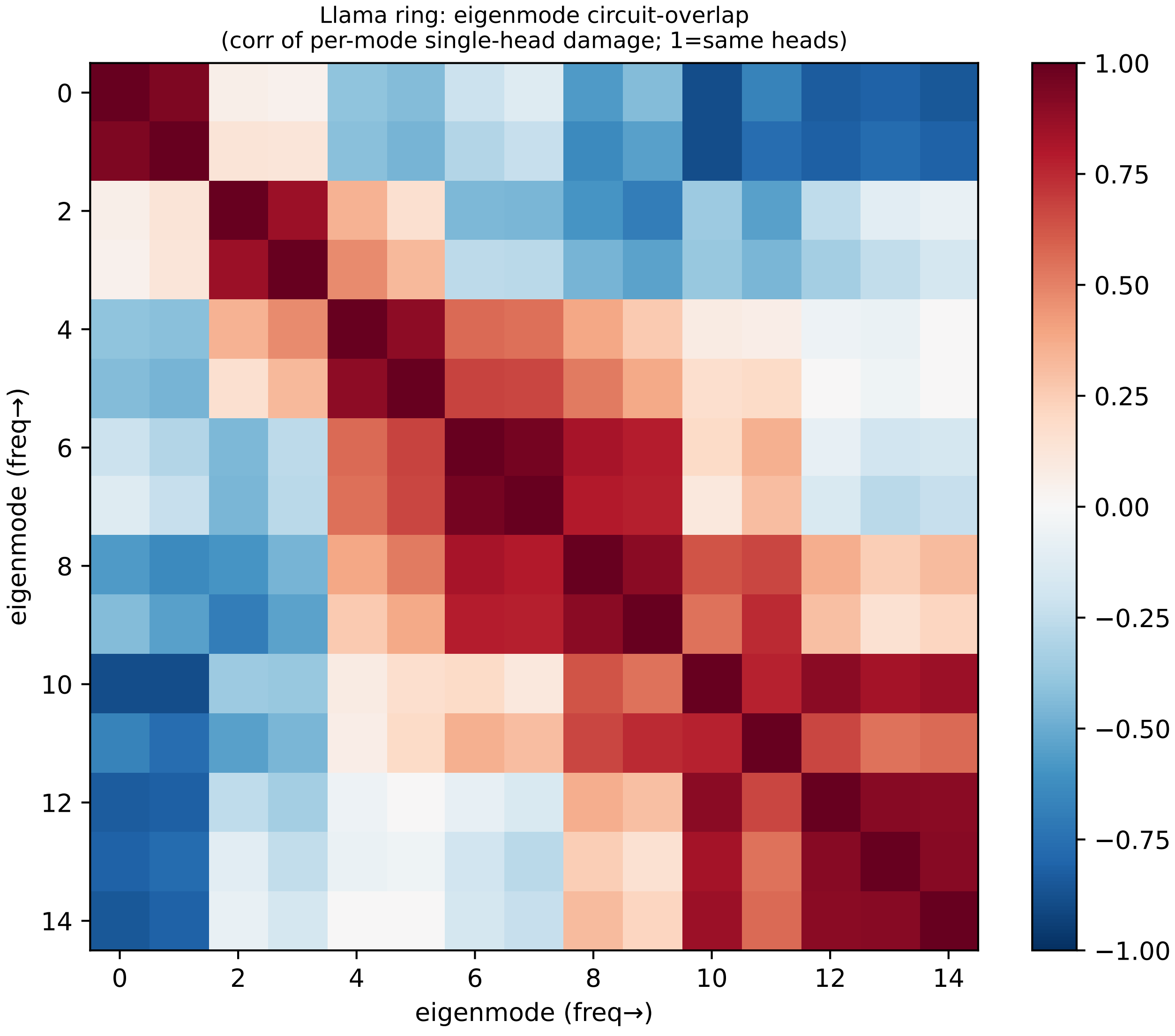
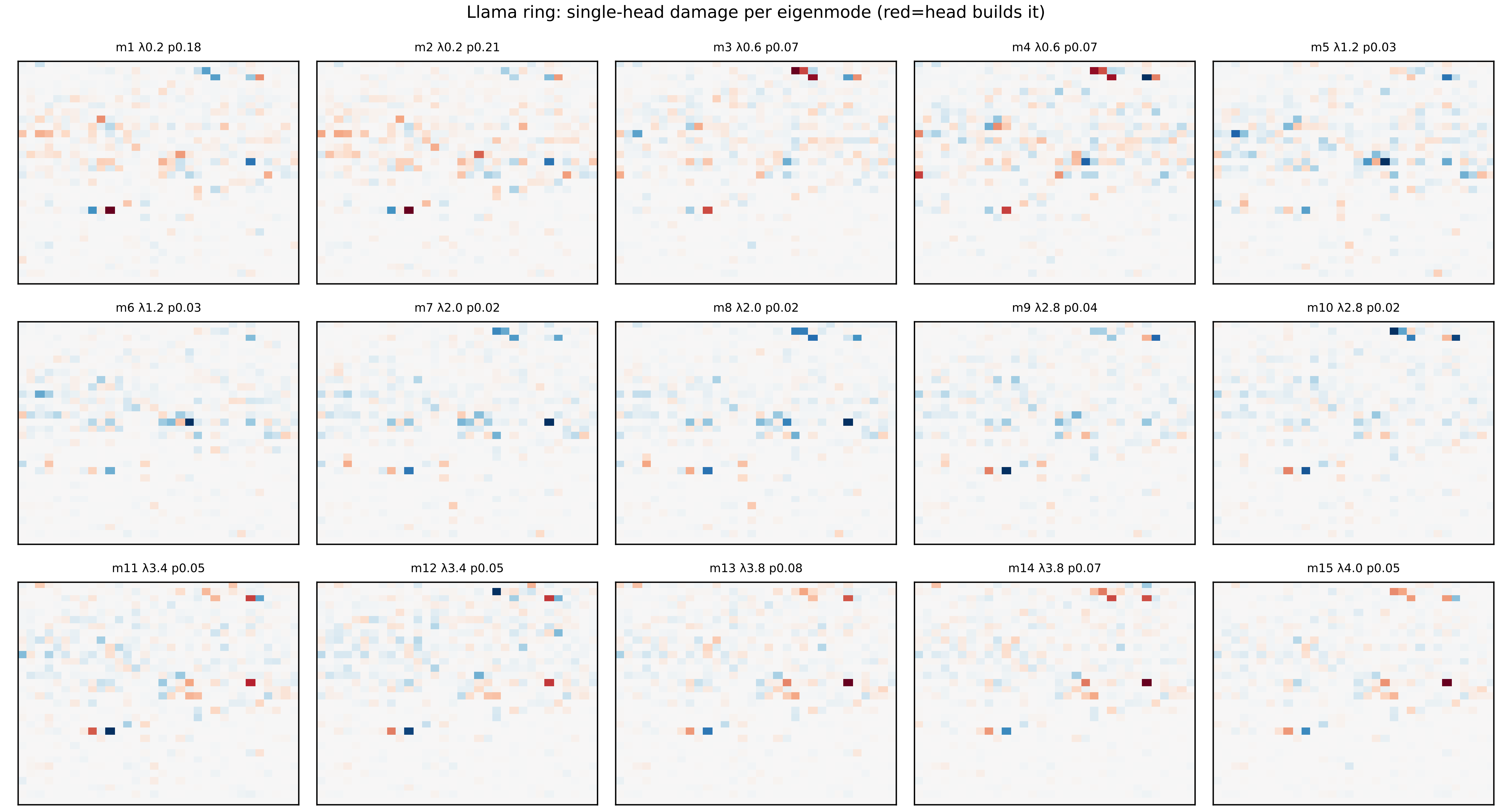

Llama ring: eigenmode circuit-overlap
(corr of per-mode single-head damage; 1=same heads)
eigenmode (freq→)
eigenmode (freq→)
0
2
4
6
8
10
12
14
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



Llama ring: single-head damage per eigenmode (red=head builds it)
m1 λ0.2 p0.18
m2 λ0.2 p0.21
m3 λ0.6 p0.07
m4 λ0.6 p0.07
m5 λ1.2 p0.03
m6 λ1.2 p0.03
m7 λ2.0 p0.02
m8 λ2.0 p0.02
m9 λ2.8 p0.04
m10 λ2.8 p0.02
m11 λ3.4 p0.05
m12 λ3.4 p0.05
m13 λ3.8 p0.08
m14 λ3.8 p0.07
m15 λ4.0 p0.05

```
Llama — ring:  which heads build which structure

CIRCUIT 1  —  x-position / y-position
    modes [2, 1] (λ 0.2, 0.2),  total power 0.39
    top heads: L21H10, L13H18, L2H27, L8H9

CIRCUIT 2  —  high-freq (no simple name) / parity/checkerboard
    modes [13, 14, 11, 15, 12] (λ 3.8, 3.8, 3.4, 4.0, 3.4),  total power 0.29
    top heads: L14H26, L2H26, L14H19, L2H22

CIRCUIT 3  —  low-freq (no simple name)
    modes [4, 3] (λ 0.6, 0.6),  total power 0.14
    top heads: L1H20, L2H22, L21H10, L1H21

Circuit 1 vs Circuit 2 cross-correlation: -0.81  (anti-correlated → compete for variance)
```